

Multi-Node Intimate Physiology System for Longitudinal Pattern Measurement, Cross-Device Correlation, and Privacy-Preserving Outputs

FIELD OF THE INVENTION

This disclosure relates to physiologic sensing devices, systems, and methods for measuring intimate-region physiology and related health context in everyday conditions. More particularly, the disclosure relates to a network of wearable and session-based sensing nodes configured to collect signals over time, derive non-anatomical and privacy-preserving metrics, and perform cross-device correlation to interpret patterns longitudinally.

SYSTEM OVERVIEW

Described herein is a modular physiologic monitoring system comprising a plurality of purpose-built sensing nodes and an associated computing system. The sensing nodes may include wearable devices for extended measurement, session-based devices for baseline acquisition, and context devices for measuring signals that improve interpretation of intimate-region physiology. The computing system may include a mobile device, a tablet, a computer, a server, or any combination thereof, and may receive data from one or more nodes via wired or wireless communication.

In various embodiments, the system includes multiple nodes configured to measure different signal modalities and to generate outputs using abstract representations rather than anatomical imaging. A first node may be configured to measure circumferential response and rigidity-related signals using one or more sensing elements distributed along a ring or partially open ring body. A second node may be configured for internal or tissue-adjacent measurement over extended wear, including pressure, temperature, hydration, moisture, impedance, motion, or other physiologic signals. A third node may be configured to measure whole-body or cardiophysiologic context, such as heart rate, heart rate variability, respiration proxies, activity, posture, stress proxies, temperature, or related signals. A fourth node may be configured to perform baseline acquisition using controlled contact, compression, or geometry-referenced routines to establish a stable individualized structural or response profile.

The computing system may transform raw sensor signals into derived features and privacy-preserving metrics such as indices, ratios, stability measures, duration measures, trend measures, and pattern curves. The computing system may further perform cross-device correlation to improve interpretation of intimate-region patterns by referencing context signals, baseline profiles, and within-user history. The system may present outputs as abstract bars, curves, scores, ranges, and confidence indicators, thereby supporting privacy-first use and longitudinal tracking without requiring explicit anatomical labeling.

SUMMARY OF KEY ADVANTAGES

In certain embodiments, the disclosed system supports (i) multi-node sensing across different physiologic contexts, (ii) longitudinal pattern measurement across days and weeks, (iii) cross-device correlation between intimate-region signals and context signals, and (iv) privacy-preserving presentation of results using non-anatomical abstractions.

INCORPORATION BY REFERENCE

Any patent, publication, or reference cited herein is incorporated by reference to the extent permitted by law.